THE IMPACT OF EXCHANGE RATE VOLATILITY ON EXPORT GROWTH: A CASE STUDY IN MALAYSIA

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1. INTRODUCTION

1.1. Background Study

Malaysia's economy is export-oriented, relying on international trade as an engine of growth and development (Hassan & Murtala, 2016; Yee et al., 2016). As an emerging open economy, Malaysia's export sector is sensitive to external shocks. Since 21 July 2005, the Malaysian Ringgit (MYR) has unpegged from the US dollar and has adopted a floating exchange rate regime, exposing the nation's exporters to significant currency volatility (Jamali et al., 2025). Therefore, understanding the relationship between exchange rate volatility and export performance is essential, as it has broad macroeconomic effects on Malaysia's economy.

The theory of how exchange rate fluctuations impact trade has developed significantly over the years. According to Purchasing Power Parity (PPP), exchange rate volatility influences export competitiveness through relative prices in international markets. Nevertheless, existing journals present a mixed picture. Studies indicate that a weakening currency strengthens export competitiveness by lowering the relative prices of exports in foreign markets (Umeaduma & Dugbartey, 2023; Hunegnaw, 2017; Wondemu & Potts, 2016). Additionally, many studies have revealed a detrimental correlation between exchange rate fluctuations and export growth (Sugiharti et al., 2020; Bahmani-Oskooee & Gelan, 2018; Upadhyaya et al., 2020). In contrast, some studies identify positive associations or determine that relationships depend on particular market characteristics and economic conditions (Muinelo-Gallo et al., 2020; Thuy & Thuy, 2019; Lee et al., 2022). Hence, it is crucial to conduct research in Malaysia to capture the complex dynamics.

This study covers the period from January 2006 to September 2025, including several significant macroeconomic events and crises (Equiti, 2025). These crises provided a rich empirical basis for investigating the relationship between currency fluctuations and export dynamics. Therefore, this study helps address the knowledge gap on Malaysia's export performance in response to exchange rate fluctuations both in the short and long run.

1.2. Problem Statement

The Malaysian economy faces significant challenges from exchange rate volatility. Figure 1 shows the fluctuations in the ringgit (MYR) against the US dollar. The volatility has created uncertainty for export-oriented firms, affecting their competitiveness, pricing strategies, and export performance (Stashaway, 2024). This is a significant challenge that requires empirical investigation to understand its mechanisms and the magnitude of its impact on Malaysian exports.

Figure 1.1: Exchange Rate (USD/MYR) 2004-2025

Source: Department of Statistics Malaysia (2025)

The fundamental problem is the ambiguous relationship between exchange rate volatility and export performance. While theories argue that depreciation can enhance export competitiveness, the Ringgit's unpredictability creates a 'risk premium' for Malaysian firms. This imposes additional costs on exporters through hedging requirements and difficulties with strategic planning, potentially neutralising any price advantage (Alagidede & Ibrahim, 2017; Barguellil et al., 2018). For a small, open economy like Malaysia, this tension between price competitiveness and volatility-related risk creates a policy dilemma that remains empirically unquantified.

Furthermore, the persistence of this problem is determined by Malaysia's volatile export performance during periods of significant currency fluctuations. For example, the 2008 global financial crisis, the global oil price crisis and recent monetary policy divergences between economies. However, empirical research on the impact of exchange rate volatility on

Malaysian export growth using ARDL or GARCH models remains limited (Hakim, 2024; Upadhyaya et al., 2020; Hook & Boon, 2017; Pino et al., 2016; Wong, 2017). Existing studies have employed broad regional analyses or focused on specific sectors, yet they lack evidence relevant to Malaysia's economic context. Hence, it justifies the study's purpose.

These research gaps necessitate empirical research that quantifies the relationship between exchange rate volatility and export growth in Malaysia, controlling for relevant macroeconomic variables and employing robust econometric techniques. Ultimately, the findings will provide policymakers and export-oriented businesses with valuable insights for formulating appropriate strategies to manage currency risk and enhance export performance.

1.3. Research Questions

I. What is the trend and magnitude of exchange rate volatility (MYR/USD) in Malaysia from 2006 to 2025?

II. What is the direction of the causal relationship between exchange rate volatility and export growth in Malaysia?

III. Does exchange rate volatility impact export growth in Malaysia in the short run?

IV. Does exchange rate volatility impact export growth in Malaysia in the long run?

1.4. Research Objectives

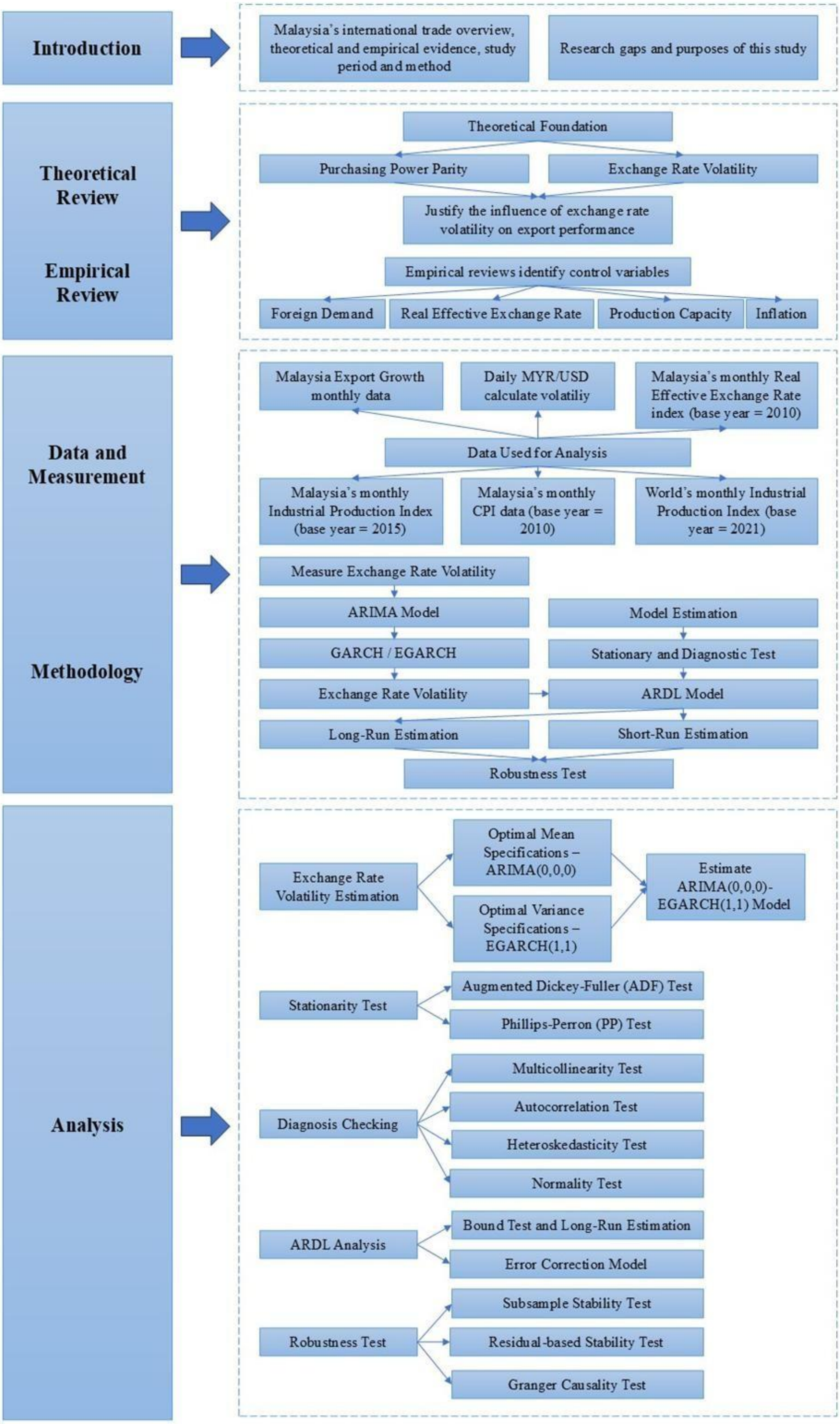
I. Estimate the conditional volatility of the MYR/USD exchange rate with the ARIMA-GARCH model.

II. Analyse the direction of causality between exchange rate volatility and export growth.

III. Examine the short-run impact of exchange rate volatility on Malaysia’s export growth using the ARDL Error Correction Model (ECM).

IV. Determine the long-run relationship between exchange rate volatility and export growth in Malaysia using the ARDL Bounds Test approach.

1.5. Visual Roadmap



2. LITERATURE REVIEW

2.1. Theoretical Review

2.1.1. Purchasing Power Parity

The Salamanca School in Spain initially proposed the Purchasing Power Parity (PPP) theory in the 16th century. Although economists have made substantial contributions to the PPP theory, it was officially established as a practical theory of exchange rates by Gustav Cassel in 1918 (Oskooee & Nasir, 2005). It suggests that exchange rate adjustment will ensure the prices of identical goods remain equivalent across countries, assuming zero transaction costs or trade barriers (Umeaduma & Dugbartey, 2025).

Absolute PPP represents the most basic and strongest form of PPP, indicating that exchange rates should adjust to equalise the prices of goods across countries (Zhang, 2024). It assumes that the real exchange rate remains constant, which is not true in reality due to transaction costs and monetary shocks (Wanzala et al., 2024). Therefore, a weaker variant of PPP, called relative PPP, suggests that exchange rate movements reflect differences in inflation rates between economies. It is considered valid when the depreciation rate of one currency against another equals the differential in inflation rates between the two economies (Sarno & Taylor, 2002).

According to Aliyu (2010), the global macroeconomic system may become significantly unstable if the nominal exchange rate consistently deviates from its PPP levels. Baldwin et al. (2005) noted that disequilibrium exchange rates are negatively associated with export performance. Domestic commodities may be mispriced in international markets when there is a discrepancy between the actual exchange rate and the expected PPP level. When volatility rises, these misalignments may cause export prices to fluctuate, potentially hindering export growth (Kusumawardani & Mubin, 2019). In contrast, some researchers suggest that exchange rate uncertainty can be fully hedged in the forward exchange market, thereby not affecting trade (Ethier, 1973; Baron, 1976). However, Viaene and de Vries (1992) contended that it may still indirectly influence trade when hedging is costly.

2.1.2. Exchange Rate Volatility and Risk Aversion Theory

The theoretical foundation for exchange rate fluctuations and risk aversion dates back to the 1970s. Early frameworks, such as the Uncovered Interest Rate Parity (UIP), were established on the assumption of complete investor risk neutrality. The theory suggests that the interest rate disparity between nations corresponds to anticipated movements in currency exchange rates (Fama, 1984). However, Kouri (1976) and Branson, Halttunen, and Masson (1977) introduced the Portfolio Balance Theory, which examined exchange rate determination from the perspective of asset market equilibrium. This theory explored the short- and long-term determinants of exchange rates, emphasising the influence of asset stocks, such as the money supply and net foreign assets, on exchange rate formation. Within this framework, investors allocate their wealth between domestic and foreign assets based on relative returns and risk tolerance. Consequently, shifts in investor preferences affect international capital flows and lead to fluctuations in exchange rates.

Upon this foundation, Branson and Henderson (1985) developed a static portfolio balance model that derived asset demand from consumers' utility-maximising behaviour. Their model analysed the interaction between money and asset markets within a general equilibrium framework, where both prices and exchange rates are endogenously determined. Later research indicated that, under a floating exchange rate system, forward exchange rates frequently serve as biased forecasts of future spot rates due to risk premiums. (Engel, 1996). However, static models fail to adequately explain short-term market fluctuations and shifts in investor sentiment, while risk premium models face empirical challenges in quantifying risk preferences. Bekaert, Hoerova, and Lo Duca (2013) investigated the relationship between the VIX index and monetary policy stance to address these issues. They revealed that accommodative monetary policy simultaneously reduces risk aversion and market uncertainty, thereby refining the theoretical framework. In practice, when investors' risk aversion intensifies, capital tends to shift from high-risk assets to safe-haven currencies (Fratzscher, 2009). These movements contribute to significant fluctuations in exchange rates, influencing global capital allocation and international trade dynamics. Thus, investors' risk-aversion influences exchange rate volatility and trade flows.

2.2. Empirical Review

2.2.1. Exchange Rate Volatility

Exchange rate volatility refers to changes in a currency's value relative to other currencies. (Melvin & Norrbin, 2023). In this hyper-globalised era, exchange rate volatility has become pronounced under floating exchange rate regimes (Tarakçı et al., 2022). Hooper and Kohlhagen (1978) argue that trading volume declines during volatile periods. In contrast, De Grauwe (1988) suggested that the influence depends on the substitution and income effects. When the income effect outweighs the substitution effect, increased exchange rate volatility boosts exports. Broll and Eckwert (1999) support this proposition when exporters can reallocate their products flexibly internationally.

Multiple studies have suggested that exchange rate volatility discourages exports (Arize, 2006; Bahmani-Oskooee & Gelan, 2018; Baum & Caglayan, 2008; Chit et al., 2010; Coric & Pugh, 2015; Dada, 2020; Fang et al., 2005; Hayakawa & Kimura, 2009; Liu & Zhang, 2025; Ozturk & Kalyoncu, 2009; Sugiharti et al., 2020). In contrast, some studies proposed a positive relationship (Chi & Cheng, 2016; Hall et al., 2010; Lee et al., 2022; Mckenzie & Brooks, 1997; Kasman & Kasman, 2006). Others suggest mixed results when analysing different countries, industries, and goods (Handoyo et al., 2023; Lal et al., 2023). Several empirical studies reach different conclusions when examining the impact in the short and long run separately. Ali et al. (2022), Bilquees et al. (2010), and Tarakçı et al. (2022) support both short- and long-term effects. Conversely, Asteriou et al. (2016) propose only a short-run impact. Moreover, Audi (2024) and Bredin et al. (2003) suggest only the long-run impact. Besides, Nishimura and Hirayama (2013) suggest that no relationship exists. Therefore, it is crucial to assess the impact of exchange rate volatility on Malaysia’s export growth.

H_1: Exchange rate volatility influences export growth in the long run.

H_2: Exchange rate volatility influences export growth in the short run.

2.2.2. Foreign Demand

Exports are fundamentally driven by foreign demand, as goods and services can only be sold abroad when there is sufficient international demand. Theories such as the Gravity model of global trade (Tinbergen, 1963) and the Export-Demand model (Kabir, 1988) have highlighted that trading partners' real income significantly affects an economy's export performance. The GDP expenditure approach suggests that foreign demand can be proxied by global consumption or world production capacity. When foreign economies expand their output, rising income levels boost consumption and, ultimately, demand for imported goods. This mechanism implies that the economic growth of trading partners can stimulate domestic exporting activities, thereby enhancing overall export growth (Barrows & Ollivier, 2020). Empirical studies have consistently supported that trading partners' growth exerts a positive influence on export performance (Ahmad et al., 2003; Arize, 1996; Bernard & Jensen, 2004; Berument et al., 2013; Bilquees et al., 2010; Bottega & Romero, 2021; Cronovich & Gazel, 1998; Hooper et al., 1998; Houthakker & Magee, 1969; Joshi & Little, 1995; Krugman, 1989; Nie & Taylor, 2013; Rahmaddi & Ichihashi, 2012; Shane et al., 2008). Therefore, foreign demand, proxied by world industrial production, is incorporated as a control variable.

2.2.3. Real Effective Exchange Rate

Real Effective Exchange Rate (REER) measures a country's external competitiveness. It is the inflation-adjusted weighted average of a country's currency relative to its major trading partners. Specifically, it reflects the relative price of a basket of foreign goods to the domestic goods basket, thus capturing the real value of the national currency (Edwards, 1989). Economic theories such as the Monetary Approach to the Balance of Payments and models of international competitiveness posit that changes in the REER significantly influence a nation's export performance (Dornbusch, 1976). A depreciation in the REER implies that a country's goods have become relatively cheaper compared to its trading partners. This enhances its export competitiveness and stimulates export growth. In contrast, an appreciation of the REER makes the economy's exports relatively more costly and can hinder export expansion (Goldstein & Khan, 1985). Additionally, existing empirical studies have confirmed that the REER exerts a significant influence on export growth and trade balances (Babatunde et al., 2017; Bahmani-Oskooee & Harvey, 2021; Darvas, 2022; Gopinath, 2023; Hidayat & Fachrudin, 2024; Irwan & Alatas, 2024; Jahazi & Arouri, 2024; Lee & Lim, 2024; Liau & Geetha, 2020; Muhammadjon & Ravshan, 2025; Ng et al., 2008). However, a few studies indicate a positive or insignificant relationship (Goda et al., 2021; Rose, 1991). Given the significance of REER in multiple studies, it is controlled in this study to account for its influence on export growth.

2.2.4. Production Capacity

Production capacity refers to the maximum output a business or production unit can generate within a fixed timeframe, based on factors such as materials, labour, and manufacturing equipment (Geršak, 2013; Amper Technologies, n.d.). The Industrial Production Index (IPI) is frequently used to measure production capacity by quantifying real output (Department of Statistics Malaysia, 2025). Neaoh et al. (2021) suggest that IPI and export growth are bidirectional, with IPI representing the supply side and export growth representing the demand side. Trade openness and increased export growth are expected to lead to industrial production growth (Yee & Bakar, 2023). Conversely, more efficient production capacity can stimulate export growth (Ahmad et al., 2018). Overall, empirical studies and official government reports have concluded that IPI is positively and significantly correlated with export-oriented growth (Maizels, 1961; Johns, 1973; Chenery & Hollis, 1980; Fung et al., 1994; Baldwin et al, 2013; Amarasinghe, 2016; Zhang & Mia, 2020; Martin et al., 2021; Normah et al., 2021; Andersen et al., 2023; Elfiana & Fachrurrozi, 2025; Ministry of Economy and Department of Statistics Malaysia, 2025). Hence, production capacity is included as a control variable.

2.2.5. Inflation

Inflation is the rise in the overall price level within an economy over time. Consumer Price Index (CPI) is widely used as a proxy for inflation. It assesses the average price variation across a consistent set of goods and services (Oner, 2010). Economic theories, notably Purchasing Power Parity (Balassa, 1964) and Cost-Based models of competitiveness (Hunt & Morgan, 1995), establish that inflation is a critical determinant of export performance. Moderate inflation depreciates currencies and makes exports more competitive internationally. However, high inflation will increase production costs and jeopardise export growth (Johnson, 1987). Moreover, the majority of existing studies indicate that inflation exerts a significant influence on export growth (Barbosa-Filho, 2006; Saygı & Emiroğlu, 2014; Momeni et al., 2014; Malau, 2016; Fachrudin & Puspitasari, 2020; Valentika et al., 2020; Jacob et al., 2021; Okpe & Ikpesu, 2021; Ilmas et al., 2022; Algul, 2024; Mamun & Larbode, 2024; Tancangco & Parcon-Santos, 2024; Zakia et al., 2024; Muhammadjon & Ravshan, 2025; Sirajuddin, 2025). Thus, inflation is incorporated as a control variable in this study.

3. DATA AND METHODOLOGY

3.1. Research Design

This is a quantitative time-series research that empirically investigates the impact of exchange rate volatility on Malaysia's export growth. A quantitative approach is selected for rigorous hypothesis testing and for identifying significant relationships among key macroeconomic variables. It adopts a two-stage modelling process: first, using the ARIMA-GARCH model to generate a conditional volatility series from daily exchange rate data; then, applying the ARDL model to estimate the short- and long-run dynamic interactions between export growth and its determinants. All analyses are performed using Python Spyder.

3.2. Data Description and Sources

The data used are monthly data from January 2006 to September 2025. The sample size is 237, which captures critical economic periods. The data details are described in Table 3.1.

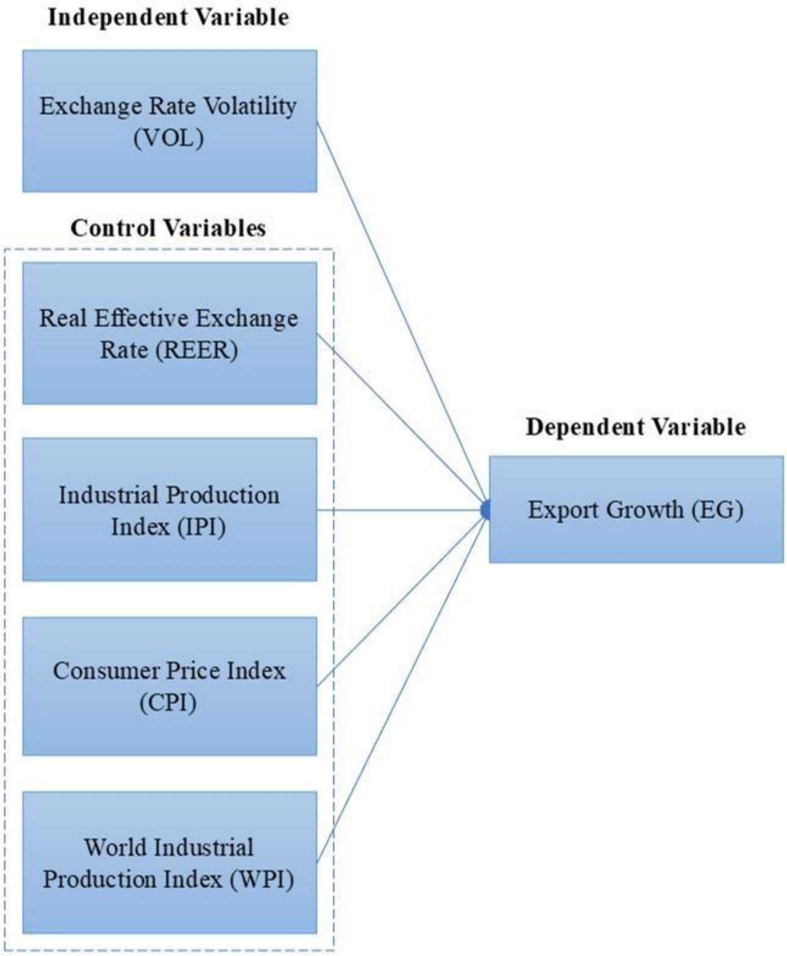
Table 3.1: Variables and Data

Variable	Symbol	Description	Data Source
Exchange Rate	EXR	MYR/USD daily data	Bank Negara Malaysia (2025)
Exchange Rate Volatility	VOL	Calculated using daily log-return of MYR/USD exchange rate and taking the monthly average	Researcher's own contribution
Export Growth	EG	Malaysia export growth, year-over-year	Department of Statistics Malaysia (2025b)
Real Effective Exchange Rate	REER	Natural log of Malaysia REER (base year = 2010)	International Monetary Fund (2025)
Industrial Production Index	IPI	Natural log of Malaysia IPI (base year = 2015)	Department of Statistics Malaysia (2025c)
Consumer Price Index	CPI	Natural log of Malaysia CPI (base year = 2010)	Department of Statistics Malaysia (2025a)
World Industrial Production Index	WIP	Natural log of World IPI (base year = 2021)	CPB Netherlands Bureau for Economic Policy Analysis (2025)

Refer to Appendix 3.2 for the descriptive statistics for the data.

3.3. Conceptual Framework

Figure 3.3: Conceptual Framework



3.4. Measurement of Exchange Rate Volatility

The ARIMA-GARCH model is employed to estimate the daily MYR/USD exchange rate volatility (Adenomon & Emmanuel, 2024; Dinga et al., 2024; Dritsaki, 2019; Ganbold et al., 2017; Perera & Rathnayaka, 2020; Qona'ah, 2023; Umar et al., 2019). The data used are the daily log returns of MYR/USD from 3 January 2006 to 30 September 2025. The ARIMA specification models the conditional mean, and the GARCH specification models the conditional variance. The details of the estimation are outlined in Appendix 3.1.

In this research, we utilise Python to determine the best-fitting model order using AIC and BIC, and to assess the significance of the asymmetry parameter. Subsequently, the Ljung-Box Q-test was used to detect for serial correlation, and the ARCH-LM test was used to detect any remaining ARCH effects. The daily conditional volatility data will be converted to monthly averages to align with the frequency of other variables in the ARDL estimation.

3.5. Model Specification

This research utilised the ARDL model to analyse how exchange rate fluctuations affect export growth, while controlling for foreign demand, the real effective exchange rate, production capacity, and the inflation rate. The ARDL model can estimate both the short- and long-run effects of variables on export growth (Appendix 3.3). It also incorporates optimal lag lengths for lagged values of the dependent and independent variables, selected based on SIC and AIC (Urgessa, 2023). According to Tarakçı et al. (2022), the ARDL model is suitable for time series data which exhibit random walk behaviour, and variables are integrated of order I(0), I(1), or a combination of both, while none of the variables should be I(2).

The Unrestricted Error Correction Model (UECM) used for the Bounds Cointegration Test allows for concurrent evaluation of both short-term and long-term coefficients. The equation is specified as follows:

$$\Delta EG_t = \alpha_0 + \sum_{i=1}^{p} \beta_{1i} \Delta EG_{t-i} + \sum_{i=0}^{q_1} \beta_{2i} \Delta VOL_{t-i} + \sum_{i=0}^{q_2} \beta_{3i} \Delta WIP_{t-i} + \sum_{i=0}^{q_3} \beta_{4i} \Delta REER_{t-i}$$

$$+ \sum_{i=0}^{q_4} \beta_{5i} \Delta IPI_{t-i} + \sum_{i=0}^{q_5} \beta_{6i} \Delta CPI_{t-i} + \lambda_1 EG_{t-1} + \lambda_2 VOL_{t-1} + \lambda_3 WIP_{t-1}$$

$$+ \lambda_4 REER_{t-1} + \lambda_5 IPI_{t-1} + \lambda_6 CPI_{t-1} + \varepsilon_t$$

Once cointegration is established, the short-run coefficients are estimated using the ECM specified as follows:

$$\Delta EG_t = \alpha_0 + \sum_{i=1}^{p} \theta_{1i} \Delta EG_{t-i} + \sum_{i=0}^{q_1} \theta_{2i} \Delta VOL_{t-i} + \sum_{i=0}^{q_2} \theta_{3i} \Delta WIP_{t-i} + \sum_{i=0}^{q_3} \theta_{4i} \Delta REER_{t-i}$$

$$+ \sum_{i=0}^{q_4} \theta_{5i} \Delta IPI_{t-i} + \sum_{i=0}^{q_5} \theta_{6i} \Delta CPI_{t-i} + \delta ECT_{t-1} + \varepsilon_t$$

Refer to Appendix 3.4 for notation explanation.

3.6. Stationarity Test

Stationarity denotes a time series characterised by a consistent mean and variance throughout time (Mohr, 2020). It is a crucial characteristic for estimating the reliable test statistics. This study employs two methods to test for stationarity. Firstly, the Augmented Dickey-Fuller (ADF) test, which incorporates lagged-difference terms to control for serial correlation in the error process, is used. Additionally, the Phillips-Perron test, which does not rely on adding lagged-difference terms, is also used for stationarity testing. For a more detailed explanation of these tests, please refer to Appendix 3.5.

3.7. Diagnostic Test

3.7.1. Multicollinearity Test

Multicollinearity occurs when the independent variables in a regression analysis are strongly correlated with one another (Wooldridge, 2019). To detect multicollinearity, this study will use the correlation matrix and the Variance Inflation Factor (VIF). VIF for each independent variable is calculated using the formula $VIF = 1/(1 - R^2_i)$, where R^2_i is obtained by regressing one variable on the remaining independent variables. A VIF value exceeding 10 indicates severe multicollinearity and warrants attention.

3.7.2. Autocorrelation Test

The most common autocorrelation diagnostics are the Durbin–Watson (DW) and the Breusch–Godfrey (BG) test. The DW test, developed by Durbin and Watson (1950), helps detect first-order autocorrelation in the residuals. However, it becomes invalid when the model includes lagged dependent variables, and it cannot detect higher-order autocorrelation. In contrast, the BG test introduced by Breusch and Godfrey (1978) is more flexible. It can identify higher-order autocorrelation and stays applicable even in the presence of lagged dependent variables. Given that this study utilises time-series data, the possibility of higher-order correlations in the residuals cannot be ignored. Therefore, the BG test is more suitable for this research.

3.7.3. Heteroskedasticity Test

Heteroskedasticity refers to non-constant variance of the residuals across the independent variables (Kroc & Zumbo, 2018). When heteroskedasticity is present, OLS estimators are inefficient, leading to incorrect standard errors. The unreliability in the standard errors undermines the validity of standard t-tests and F-tests, potentially leading to inaccurate interpretations of the statistical significance of the variables. The Breusch-Pagan (BP) test is used to detect heteroskedasticity (Breusch & Godfrey, 1978). If the p-value is insignificant, the model is homoskedastic.

3.7.4. Normality Test

The estimation procedure for a normality test assesses whether the model's residuals are normally distributed. Visual inspections of residuals in a histogram and Quantile-Quantile (QQ) plot provide an intuitive way to determine the normality assumption (Wilk & Gnanadesikan, 1968). The histogram should visually show a bell-shaped, symmetrically centred curve if the residuals are normally distributed. For a QQ plot, the plotted points must lie closely along a straight line at 45 degrees to signal normality. Additionally, the Jarque-Bera (JB) Test will also be used to test for normality.

3.8. Robustness Check

3.8.1. Subsample Stability Test

The Subsample Stability Test splits the entire dataset into subsamples. This is often done by identifying a known structural break, such as a significant policy change, an economic crisis, or a regime shift, or by splitting the sample into two or three equal parts (Andrews, 2003). Then re-estimate with the same ARDL/ECM model for each subsample separately. After that, compare the long-run and short-run coefficients, as well as the ECM speed of adjustment, across the subsamples. If the coefficients across the subsamples are statistically similar, the model is considered robust.

3.8.2. Residual-based Stability Test

Residual-based stability tests, such as the CUSUM and CUSUMSQ tests, assess whether the estimated regression coefficients remain stable over time. The procedure begins by estimating the model and generating recursive residuals by repeatedly re-estimating the regression as the sample is gradually expanded (Brown et al., 1975). These recursive residuals form the basis of the test statistics. The CUSUM test calculates the cumulative sum of standardised recursive residuals, while the CUSUMSQ test uses the cumulative sum of squared residuals. Each test statistic is then plotted against time. The model's parameters are considered robust if the plotted line is within the critical bounds.

3.8.3. Granger Causality Test

The Granger causality test evaluates whether past values of one variable can help predict another by comparing a restricted and an unrestricted forecasting model (Granger, 1969). The time-series variables must be stationary before testing. If the variables are non-stationary but cointegrated, a VECM-based Granger causality framework is used instead of a standard VAR. The next step is to estimate a VAR or VECM model with the optimal lag structure. The null hypothesis states that historical values of one variable cannot predict another variable. Besides, the test is implemented by evaluating whether the lagged coefficients of the predictor variable jointly equal zero by using a Wald or F-test. If the result is statistically significant, it indicates that the past values provide statistically significant predictive information for the other.

4. RESULTS AND DISCUSSION

4.1. Exchange Rate Volatility Estimation

Based on the AICs, the best-fitting ARIMA model is ARIMA(0,0,0), and the EGARCH(1,1) is superior for estimating MYR/USD volatility dynamics. The Student's t-distribution is used as it has a lower AIC than assuming a normal distribution. The output for the EGARCH(1,1), mean, and volatility models is shown in Table 4.1.3 in Appendix 4.1. The dependent variable Scaled_Return is the exchange rate log returns multiplied by 100 for scaling. Log-likelihood of 2241.75 indicates successful convergence. R-squared of nearly zero confirms the Random Walk Hypothesis in the exchange rate. The mean model suggests that the daily average exchange rate return does not exhibit a long-term trend. For the volatility model, the ARCH term confirms volatility clustering. Additionally, the asymmetry term suggests that there is no leverage effect and the MYR/USD exchange rate reacts symmetrically. Furthermore, the GARCH term indicates high persistence of volatility shocks.

For more details on the volatility model diagnostic checking, descriptive statistics and trend analysis, kindly refer to Appendix 4.1.

4.2. ARDL Bound Test and Long-Run Estimation Results

Upon completing the stationarity test (Appendix 4.2) and the diagnostic test (Appendix 4.3), we performed the ARDL bound test. The ARDL bound test yielded an F-statistic of 6.9242, with upper bounds of 4.68 (1%) and 3.79 (5%). The F-statistic significantly exceeds the upper bound value at the 1% and 5% significance levels (Pesaran et al., 2001). Hence, we reject the null hypothesis and conclude that there is a stable long-run relationship among export growth, exchange rate volatility, and the selected control variables in Malaysia.

The long-run coefficient results are presented in Table 4.1 below. The primary variable of interest, exchange rate volatility (VOL), has a negative coefficient of -64.7091. The relationship is statistically significant at the 10% level (p = 0.0699). Although it exceeds the 5% threshold, given the volatility of the sample period, it provides moderate evidence that exchange rate volatility exerts negative pressure on exports. In the long run, exchange rate volatility harms exports.

As for control variables, REER and IPI have negative coefficients, while WPI and CPI have positive coefficients. However, none are statistically significant in the long run, not even at the

10% significance level. The insignificance of control variables, despite strong cointegration in the ARDL bound test, may be due to high multicollinearity (Appendix 4.3). It inflates standard errors and makes t-statistics unreliable. Although the individual t-statistics for the control variables are low, the significant bound test F-statistic suggests that the variables are jointly substantial in their predictive power for export growth. Thus, multicollinearity exists but does not fatally affect the model's explanatory power.

Table 4.1: Long-Run Coefficients

Variable	Coefficient	Standard Error	t-stat	p-value
VOL	-64.7091	35.5273	-1.8214	0.0699
L_REER	-71.7734	51.8610	-1.3840	0.1678
L_WPI	98.8305	104.1554	0.9489	0.3438
L_CPI	2.7549	61.2830	0.0450	0.9642
L_IPI	-69.8590	71.5348	-0.9766	0.3299

4.3. Short-Run Estimation Results

The short-run relationships and speed of adjustment towards equilibrium were estimated using the ECM and summarised in Appendix 4.4. The coefficient of the error-correction term is labelled ECT_Lag1. The coefficient is -0.2918, and is highly significant (p = 0.000). Theoretically, a negative coefficient satisfies the requirement for convergence, and the magnitude of the coefficient indicates a moderate speed of adjustment. In context, a 29.2% deviation from the long-run equilibrium due to short-run impacts can be corrected within one month. This implies that it takes approximately three to four months for the system to fully return to its equilibrium path following a disturbance.

As for short-run relationships, the impact of exchange rate volatility (VOL) remains negative with a coefficient of -28.1536. But it is statistically insignificant (p = 0.142). This suggests that volatility does not cause immediate statistically discernible change in export growth in the very short term. In contrast, some control variables show significant short-run effects. CPI and IPI have coefficients of 296.3207 (p = 0.044) and 30.0177 (p = 0.034), respectively, exhibiting a positive short-run relationship. This suggests that domestic supply-side factors have a greater impact on monthly export fluctuations than exchange rate volatility. WPI is nearly significant at the 10% level (p = 0.108) with a coefficient of 103.9833, and REER is insignificant.

4.4. Robustness Tests

4.4.1. Subsample Stability Test

The subsample stability test splits the data into two periods: 2006-2015 and 2016-2025. From Table 4.2, results show that, except for VOL, all other coefficients have a consistent sign across the two periods. The coefficient of VOL for subsample 1 is positive (21.7499), but negative (-79.5359) for subsample 2. This distinct shift in coefficients between the two periods suggests that the relationship between exchange rate volatility and export growth is regime-dependent, with a negative impact becoming prevalent in the current period. This is likely due to the compounding effects of the 2015 1MDB scandal crisis, 2020 COVID-19 pandemic, and the recent aggressive US monetary tightening (Hui, 2021; Shahrier, 2022; Soo et al., 2025). This confirms that the relationship has evolved structurally over the last decade due to changes in global macroeconomic conditions.

Table 4.2: Subsample Stability Check

Variable	2006-2015	2016-2025
VOL	21.7499	-79.5359
L_REER	-237.3910	-86.4930
L_WPI	200.1337	105.3915
L_CPI	-21.0535	-129.1289
L_IPI	-103.8825	-14.5235

4.4.2. Residual-based Stability Test

According to Figure 4.3, the CUSUM statistic stays within the critical limits at the 5% significance level. This suggests that the model's coefficients are generally stable in terms of the mean, with only two significant spikes around observations 30 and 170 on the x-axis, corresponding to mid-2008 and early 2020. These dates are in accordance with the 2008 global financial crisis and the 2020 COVID-19 pandemic.

In Figure 4.4, the CUSUMSQ plot crosses the critical bounds. This suggests instability in the variance of the regression error. The failure occurs most significantly between 0.13 and 0.74 on the x-axis, consistent with significant structural breaks identified in the trend analysis (Appendix 4.1), mainly the 2008 financial crisis and the COVID-19 pandemic.

The two stability tests offer a nuanced insight into the model. The stability of the CUSUM plot indicates that the coefficients are robust and the relationships between variables have remained consistent over the study period. On the other hand, the instability in the CUSUMSQ plot indicates that the predicted errors are not constant and have changed over time. It reflects the model's weaknesses in failing to maintain constant variance during extreme events. However, this clustering is expected, given that the heightened global uncertainty is a common phenomenon for financial time-series data.

In conclusion, the robustness check identified a dynamic relationship, and the model effectively captures the long-run relationship. The influence of exchange rate volatility on export performance is regime-dependent and prone to instability during a global crisis.

Figure 4.3: CUSUM Plot

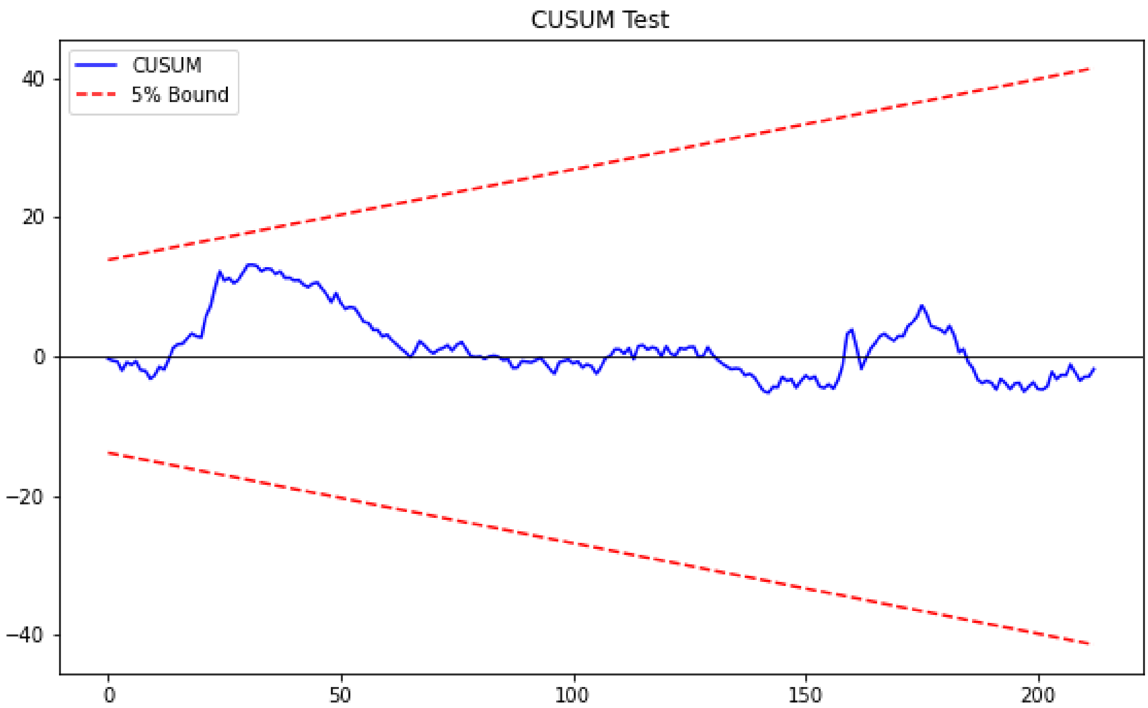
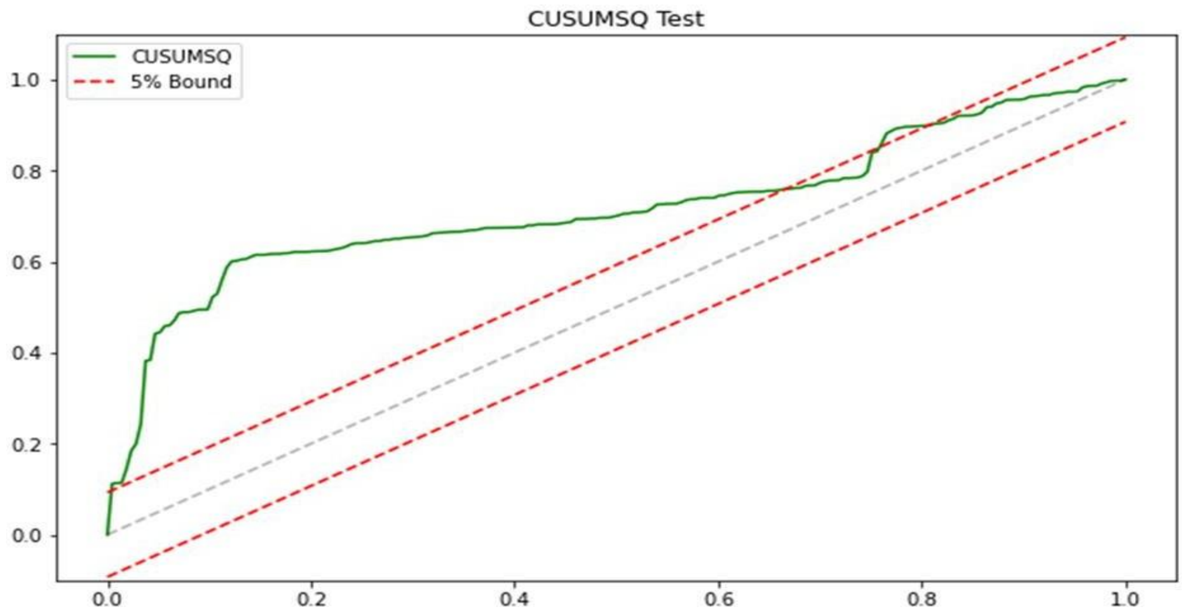


Figure 4.4: CUSUMSQ Plot



4.5. Granger Causality Test

The Granger causality test results summarised in Table 4.5 indicate that the VOL-to-EG direction is significant at the 10% significance level (p = 0.0656). This suggests a weak unidirectional causality running from exchange rate volatility to export growth. As for WPI and CPI, they exhibit significant causal effects on EG, with p-values of 0.0041 and 0.0143, respectively. This reinforces previous findings that global demand and domestic cost conditions (supply-side) are the fundamental drivers of Malaysia's export performance. Additionally, neither the REER nor the IPI showed significant causality for exports.

In short, the causality test suggests that while exchange rate volatility is the secondary disruptor of exports, fundamental demand and supply side structures are the root driver of Malaysia's export performance.

Table 4.5: Granger Causality Test

Direction	p-value	Result
VOL → EG	0.0656	Weak Cause
L_REER → EG	0.4882	No Cause
L_WPI → EG	0.0041	Causal
L_CPI → EG	0.0143	Causal
L_IPI → EG	0.4533	No Cause

4.6. Discussion

The unrestricted ARDL model exhibits a moderate-to-high goodness-of-fit (R^2 = 0.6692) and is statistically significant overall (Appendix 4.5). Moreover, the ARDL bound test indicates a valid long-run relationship. Exchange rate volatility cointegrates with Malaysia's export performance at the 10% significance level in the long run, aligning with past studies (Audi, 2024; Bredin et al., 2003). Hence, H_1 is supported. The other variables are insignificant, possibly due to high multicollinearity (Appendix 4.3). However, we do not implement remedial measures for two reasons. First, dropping variables would lead to model misspecification, failing to capture the whole structure of the theory and thereby invalidating the model. Besides, the goal of the ARDL model is to derive the robust long-run coefficients and the ECT. Since multicollinearity only affects the standard errors of the individual short-run lag terms, the long-run coefficient estimates and the ECT remain reliable and unbiased.

The short-run estimation indicates that MYR/USD exchange rate volatility does not affect Malaysia's export performance in the short term. Hence, H_2 is rejected. The results support the short-run impact of the error-correction term, past export growth, the Consumer Price Index, and the Industrial Production Index on predicting export growth. Additionally, the world Industrial Production Index is potentially a short-run determinant of export growth, being nearly significant at the 10% level.

The robustness tests identified some unexpected characteristics. Firstly, the relationship between exchange rate volatility and export growth is dynamic. It was positive before 2015 but negative afterwards as global macroeconomic conditions changed. Secondly, the model effectively captures the long-run relationship, but the relationship is regime-dependent. The relationship is more unstable during shocks. The Granger causality test indicates that exchange rate volatility is a secondary driver of exports, while the World Industrial Production Index and the Consumer Price Index are the primary drivers. This suggests that Malaysian exports are highly dependent on global demand conditions and supply-side inflationary pressure.

Moreover, the finding that exchange rate volatility exerts a long run negative impact on Malaysia's export growth offers an important refinement to international economics theory. It aligns with De Grauwe's (1988) classical framework, which posits that risk-averse firms reduce trade volumes in the presence of exchange rate uncertainty. However, this study provides more specific empirical insight, showing that the risk-aversion mechanism significantly dominates any potential offsetting effects, such as risk-seeking behaviour or widespread effective hedging in Malaysia. For Malaysian exporters, the threat of profit loss outweighs the incentive to expand sales, reinforcing risk aversion as the prevailing behavioural assumption for this market. In contrast, the short-run insignificance of exchange rate volatility mirrors the mixed empirical findings worldwide. This neutrality may arise because large exporters often hedge short-term currency risks. Exporters may also delay major adjustments until the volatility trend appears persistent, resulting in a statistically insignificant short-run effect. Overall, the findings highlight the need for policymakers in emerging markets to manage exchange rate risk actively. Rather than framing the issue as "float versus fix", the evidence suggests that any policy reducing persistent volatility can support stronger export performance and macroeconomic stability. At the firm level, since short-term volatility is unavoidable under a flexible regime, exporters should enhance non-price competitiveness, such as quality, branding, and product differentiation, to reduce sensitivity to currency fluctuations. This study thus helps address inconsistent volatility, showing that higher inherent volatility and less sophisticated hedging structures tend to reinforce the theoretically predicted negative effect.

The findings hold significant practical implications. The ARIMA-GARCH model estimates the conditional volatility of the MYR/USD exchange rate and indicates that exchange rate fluctuations in Malaysia exhibited significant clustering characteristics during the period. This is consistent with previous studies on small open economies, which demonstrate that, while exchange rate fluctuations persist, they do not exhibit a trend toward structural expansion. Furthermore, the results of the short-term error-correction model suggest that immediate fluctuations in exports are more likely to be driven by changes in production costs, supply constraints, and capacity utilisation, rather than short-term movements in the transaction currency's exchange rate. Therefore, policies to stimulate exports immediately should focus on the supply side, while initiatives to sustain export growth should focus on lowering exchange rate volatility. Moreover, this study found significant multicollinearity among domestic macroeconomic variables. This implies that policy tools cannot operate in isolation as interventions targeting one variable inevitably generate spillover effects to another. Therefore, macro policy design should adopt a coordinated approach.

Despite providing valuable insights, our research still has some limitations. Firstly, the variables in our study exhibit significant multicollinearity, which attenuates the significance of the key long-term coefficients in our model. Although multicollinearity increases the standard error, the Ordinary Least Squares (OLS) estimator remains the Best Linear Unbiased Estimator

(BLUE) within the ARDL framework, and the overall model remains robust. Secondly, aggregate macro export data may mask the heterogeneity across industries. Different export categories, such as manufacturing, commodities, or services, may have varying sensitivities to exchange rate fluctuations. Due to the complexity of the discussion, our study did not make these distinctions. Thirdly, the linear ARDL model may not fully capture the potential nonlinear and asymmetric effects of exchange rate fluctuations on exports across different time periods. A separate time-horizon analysis should be conducted to capture the evolving relationship between the variables.

5. CONCLUSION AND RECOMMENDATION

In conclusion, this study investigated the impact of MYR/USD volatility on Malaysia's export growth using data from January 2006 to September 2025. The result indicates that exchange rate volatility has no short run impact on export growth significantly. This is justified by exporters' ability to implement hedging strategies to eliminate short-term exchange rate risk. In fact, short-term export performance is mainly driven by economic factors related to supply and demand. In the long run, the exchange rate volatility influence export growth at a 10% significance level. It suggests that as volatility persists, risk-aversion behaviour becomes prevalent, harming exports. The robustness test indicates a structural break in the post-2015 period, in which the relationship shifted from positive to negative. Additionally, the model exhibits better capability to capture the relationship in normal scenarios, but becomes unstable during a crisis.

The findings suggest that policymakers should emphasise supply-side factors when targeting short-run export growth. In contrast, they must focus on exchange rate stability when aiming to promote long-term export growth. Additionally, the policies must be formulated with a comprehensive perspective, given the significant spillover effects on the economy. On the other hand, exporting firms should enhance their non-price competitiveness to minimise the damage from the loss of pricing advantage when exchange rate volatility is high.

As for handling the limitations, future research should address multicollinearity by employing Principal Component Analysis (PCA) to develop a composite index of economic activity. Alternatively, Dynamic Factor Models help isolate common trends among variables and enable more precise estimation. Besides, future research can conduct a sectoral analysis that categorises different goods to capture the varying relationships. For example, differentiating between manufactured goods and commodity exports can provide better estimates and more granular policy insights. Lastly, future research should employ the Non-linear ARDL (NARDL) model to capture the potential asymmetries in the model.

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APPENDIX

Appendix 3.1

The mean equation for the ARIMA(p,d,q) is:

$$\phi(L)(1-L)^d y_t = c + \theta(L)\varepsilon_t$$

where: y_t = exchange rate at time t

L = lag operator $L^k y_t = y_t - k$

d = differencing order

c = constant

ε_t = error term at time t

When d = 1, the formula in expanded form is:

$$\Delta y_t = c + \sum_{i=1}^{p} \phi_i \Delta y_{t-i} + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j} + \varepsilon_t$$

The error term ε_t in this equation is the input for the volatility model used to test for ARCH effects. We assume $\varepsilon_t = \sigma_t z_t$ that z_t follows a standard normal distribution and σ_t is the conditional variance.

After the ARIMA model, both the GARCH and EGARCH models are employed to estimate conditional volatility, accounting for volatility clustering and potential asymmetric leverage effects. The variance equations for the standard GARCH and EGARCH models are as follows:

Standard GARCH(p,q): $\sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2$

where: σ_t^2 = Conditional volatility for day t

ω = intercept

ε_{t-i}^2 = Past squared residuals

σ_{t-j}^2 = Past volatility

with Constraint: $\sum_{i=1}^{q} \alpha_i + \sum_{j=1}^{p} \beta_j < 1$ to ensure stationarity

EGARCH(p,q): $$\ln(\sigma_t^2) = \omega + \sum_{j=1}^{p} \beta_j \ln(\sigma_{t-j}^2) + \sum_{i=1}^{q} \left(\alpha_i \left|\frac{s_{t-i}}{\sigma_{t-i}}\right| + \gamma_i \frac{s_{t-i}}{\sigma_{t-i}}\right)$$

where: $\ln(\sigma_t^2)$ = Log conditional volatility

ω = Intercept

α_i = Effect size of the shock

γ_i = Asymmetry parameter (Asymmetry when $\gamma_i \neq 0$)

Appendix 3.2: Descriptive Statistics

Table 3.2: Descriptive Statistics

	EG	EXR	REER	WPI	CPI	IPI
Mean	7.1926	3.8197	92.6389	89.2034	106.1078	95.0899
Std Error	0.7583	0.0075	0.4223	0.5641	0.9559	1.2427
Median	6.509	3.923	94.803	89.09	105.5	91.067
Mode	12.217	4.1845	94.214	74.401	81.5	62.821
Std Dev	13.3296	0.5233	7.4238	9.9164	16.8026	21.8453
Variance	177.6785	0.2739	55.1132	98.3354	282.3258	477.2159
Kurtosis	1.3091	-1.3519	-1.1941	-1.1169	-1.2773	-0.9278
Skewness	0.1554	-0.0408	29.574	-0.1023	0.0114	0.2076
Range	93.153	1.855	77.159	34.66	54.9	85.865
Minimum	-30.421	2.9385	106.706	71.836	80.3	55.146
Maximum	62.732	4.7935	28625.418	106.496	135.2	141.011
Count	309	4887	309	309	309	309

Appendix 3.3: Long-Run and Short-Run Estimation

The autoregressive distributed lag (ARDL) bounds testing approach is widely used to assess the presence of a long-term relationship among variables. Unlike traditional cointegration methods such as the Engle–Granger or Johansen procedures, the ARDL bounds test does not require all variables to share the same order of integration; it is valid as long as none of the variables is integrated of order I(2) (Pesaran et al., 2001). This flexibility makes the ARDL framework suitable for small-sample settings and for models that include variables with different degrees of stationarity. The basic model of ARDL is as follows:

$$\Delta y_t = \alpha_0 + \sum^{p} \phi_i \Delta y_{t-i} + \sum^{q} \theta_j \Delta y_{t-j} + \lambda_1 y_{t-1} + \lambda_2 x_{t-1} + \varepsilon_t$$

$i=1$ $j=1$

The bounds test is conducted by performing an F-test on the joint significance of the lagged level variables:

$$H_0: \lambda_1 = \lambda_2 = 0 \qquad \text{(No long-run relationship exists)}$$

$$H_1: \lambda_1 \neq 0, \lambda_2 \neq 0 \qquad \text{(Cointegration exists)}$$

Pesaran et al. (2001) provide two sets of asymptotic critical values, one for the lower bound (I(0)), assuming all variables are stationary. The other for the upper bound (I(1)), considering all variables are first-difference stationary. The decision rule is that if the computed F-statistic exceeds the upper bound, the null hypothesis of no cointegration is rejected. Additionally, tf the F-statistic falls below the lower bound, the null cannot be rejected. Lastly, if the F-statistic falls within the bounds, the test is inconclusive.

The long-run coefficients are calculated from the ARDL model's estimated coefficients. The long-run model can be represented as:

$$Y_t = \alpha_0 + \sum_{i=1}^{k} \beta_i X_{it} + \epsilon_t$$

The long-run estimation coefficients show the permanent impact of a unit change in an independent variable to the dependent variable. For example, a long-run coefficient (β) of 0.8 for X means that a permanent 1 unit increase in X will lead to 0.8 unit increase in the long-run equilibrium level of $\boldsymbol{Y}$. Besides, the short-run estimation is derived from the reparameterized ECM and contains coefficients associated with the differenced terms. The short-run coefficients are interpreted as standard regression coefficients in a differenced model, representing the immediate effect of a 1 unit change in the independent variable on the current change in the dependent variable. Critically, inside the ECM, it includes the ECT, which represents the lagged residual from the long-run cointegrating equation. The ECT coefficient must be negative and statistically significant for cointegration to hold. Its magnitude represents the speed of adjustment, which is the proportion of the previous period's disequilibrium that is corrected in the current period to restore the long-run equilibrium. For example, a coefficient (δ) of -0.45 means that 45% of the deviation from the long-run equilibrium in the previous period is corrected in the current period (Pesaran et al., 2001).

Appendix 3.4

$$\Delta EG_t = \alpha_0 + \sum_{i=1}^{p} \beta_{1i}\Delta EG_{t-i} + \sum_{i=0}^{q1} \beta_{2i}\Delta VOL_{t-i} + \sum_{i=0}^{q2} \beta_{3i}\Delta WIP_{t-i} + \sum_{i=0}^{q3} \beta_{4i}\Delta REER_{t-i}$$

$$+ \sum_{i=0}^{q4} \beta_{5i}\Delta IPI_{t-i} + \sum_{i=0}^{q5} \beta_{6i}\Delta CPI_{t-i} + \lambda_1 EG_{t-1} + \lambda_2 VOL_{t-1} + \lambda_3 WIP_{t-1}$$

$$+\lambda_4 REER_{t-1} + \lambda_5 IPI_{t-1} + \lambda_6 CPI_{t-1} + \varepsilon_t$$

Where p = number of lags for the export growth

q_i = number of lags of the independent and each control variable

β_i = short-run coefficients

λ = long-run coefficients

$$\Delta EG_t = \alpha_0 + \sum_{i=1}^{p} \theta_{1i}\Delta EG_{t-i} + \sum_{i=0}^{q1} \theta_{2i}\Delta VOL_{t-i} + \sum_{i=0}^{q2} \theta_{3i}\Delta WIP_{t-i} + \sum_{i=0}^{q3} \theta_{4i}\Delta REER_{t-i}$$

$$+ \sum_{i=0}^{q4} \theta_{5i}\Delta IPI_{t-i} + \sum_{i=0}^{q5} \theta_{6i}\Delta CPI_{t-i} + \delta ECT_{t-1} + \varepsilon_t$$

Where θ = short-run coefficients

ECT_{t-1} = lagged value of the residual term obtained from the cointegration equation

δ = speed of adjustment towards the equilibrium

Appendix 3.5: Stationarity test details

The Augmented Dickey-Fuller (ADF) test follows the Dickey-Fuller (1979) framework and extends it by incorporating lagged-difference terms to control for serial correlation in the error process. It is one of the most commonly used methods for detecting unit roots in time series. The presence of unit roots is a key indicator of non-stationarity in time series.

The standard regression model of the ADF test can be written as:

$$\Delta x_t = \alpha + \beta t + \gamma x_{t-1} + \sum_{i=1}^{p} \delta_i \Delta x_{t-i} + \varepsilon_t$$

$\Delta x_t = x_t - x_{t-1}$

p = number of lagged difference terms

γ = coefficient of interest, where $\gamma = 0$ means non-stationarity, while $\gamma < 0$ means stationarity

The result depends on the statistical significance of γ, where:

$$H_0: \gamma = 0 H_1: \gamma < 0$$

If p-value < 0.05, reject H_0 and conclude that the series is stationary

If p-value > 0.05, fail to reject H_0 and conclude that the series is non-stationary

In time series analysis, selecting an appropriate lag length is crucial when building autoregressive or vector autoregressive models. If too few lag terms are included, the model may fail to capture the data's dynamic structure, leading to underfitting, biased parameter estimates, and reduced predictive accuracy. Conversely, using too many lag terms can cause the model to overfit random noise, thereby weakening its generalisation ability and predictive stability (Liew, 2004; Petrică et al., 2017). In this study, the optimal lag length is determined using the Akaike Information Criterion (AIC) and the Schwarz Information Criterion (SIC).

The two criteria are defined as below:

Akaike Information Criterion (AIC):

$$AIC = -2L + 2k$$

Schwarz Information Criterion(SIC):

$$SIC = -2L + kln(n)$$

L = log-likelihood of the estimated model

k = number of estimated parameters (including lag terms)

n = sample size

Because the penalty term (2k) increases linearly with the number of parameters, AIC tends to favour models with more lags to improve in-sample fit (Akaike, 1973). In contrast, since $\ln(n)$ it typically exceeds 2 for moderate or large samples, SIC imposes a more substantial penalty on model complexity than AIC and thus selects more parsimonious lag structures. In empirical applications, AIC is often preferred when predictive accuracy is the primary objective, whereas SIC is more suitable when model simplicity and robustness are prioritised. The optimal lag length is determined by comparing the criterion values across candidate models and selecting the lag order that minimises either AIC or SIC.

Another stationarity test employed is the Phillips-Perron (PP) test. Unlike the ADF test, the PP test does not rely on adding lagged-difference terms to correct for serial correlation. Instead, it uses a non-parametric adjustment to modify the t-statistic, making it more robust to heteroskedasticity and mild autocorrelation in the error term (Petrică et al., 2017). Overall, although the correction methods used in the PP and ADF tests differ, they typically yield similar empirical results.

In the case of the PP test, the regression equation itself is the standard Dickey-Fuller form:

$$\Delta x_t = \alpha + \beta t + \gamma x_{t-1} + \varepsilon_t$$

To account for autocorrelation, the PP test adjusts the conventional $t_\alpha$statistic and produces the corrected statistic $\underset{-}{t_\alpha}$defined as:

$$\underset{-}{t_\alpha} = t_\alpha \left(\frac{\gamma_0}{f_0}\right)^{\frac{1}{2}} - \frac{T(f_0-\gamma_0)(se(\hat{\alpha}))}{2 f_0^{\frac{1}{2}} s}$$

s = the standard error of the test regression

γ_0 = represents a consistent estimate of the error variance

f_0 = the estimated residual spectrum at frequency zero

Appendix 4.1

Determination of Optimal Mean and Variance Specifications

The AICs for the ARIMA models with different orders are summarised in Table 4.1.1. The result determines that the best-fitting ARIMA model is ARIMA(0,0,0). This indicates that the conditional mean of MYR/USD returns is a white-noise process, and that the exchange rate's dynamic structure is reflected in the returns' variance (volatility). Therefore, GARCH models are essential to assess volatility clustering.

Table 4.1.1: Summary of the AIC of different ARIMA models

ARIMA Models	AIC
ARIMA(0,0,0)	-2980.713
ARIMA(1,0,0)	-2978.714
ARIMA(0,0,1)	-2978.714
ARIMA(1,0,1)	-2976.713
ARIMA(0,0,0) intercept	-2978.815

The comparison of the GARCH and EGARCH models under the normal distribution assumption is presented in Table 4.1.2. The EGARCH(1,1) model shows lower AIC and BIC, and higher log-likelihood. Hence, it is superior for estimating MYR/USD volatility dynamics.

Table 4.1.1: GARCH and EGARCH comparison

Models	AIC	BIC	Log-likelihood
GARCH(1,1)	-4016.1037	-3990.1264	2012.0519
EGARCH(1,1)	-4049.6017	-4017.1300	2029.8008

Estimation of the ARIMA(0,0,0)-EGARCH(1,1) Model

The Student's t-distribution is used to estimate the model, as financial time-series data commonly have fatter tails. The AIC also shows that Student's t-distribution (AIC = -4471.50) is superior to the normal distribution (AIC = -4049.60). The output for the EGARCH(1,1), mean, and volatility models is shown in Table 4.1.3. The dependent variable Scaled_Return is the log returns multiplied by 100 for scaling. Log-likelihood of 2241.75 indicates successful convergence. R-squared of nearly zero confirms the Random Walk Hypothesis in the exchange rate. The mean model suggests that the daily average exchange rate return is statistically insignificant, implying the absence of a long-term trend in daily returns. For the volatility model, the ARCH term alpha[1] is highly significant ($p < 0.01$). It confirms volatility clustering, in which yesterday's shock directly affects today's volatility. The asymmetry term gamma[1] is insignificant (p = 0.430), suggesting that there is no leverage effect and the MYR/USD exchange rate reacts symmetrically. The GARCH term beta[1] is highly significant ($p < 0.01$) with a coefficient close to 1. It indicates high persistence, meaning a volatility shock has a long-lasting effect on the exchange rate.

Table 4.1.3: EGARCH, mean, and volatility models results

Constant Mean – EGARCH Model Results				
Dependent Variable:	Scaled_Return	R-squared:	0.000	
Mean Model:	Constant Mean	Adj. R-squared:	0.000	
Volatility Model:	EGARCH	Log-likelihood:	2241.75	
Distribution:	Standardised Student's t	AIC:	-4471.50	
Method:	Maximum Likelihood	BIC:	-4432.59	
		Observations:	4844	
		Df Residuals:	4843	
		Df Model:	1	
Mean Model				
	Coefficient	Standard Error	t-statistic	p-value
mu (Constant)	-0.0014	0.0018	0.774	0.439
Volatility Model				
	Coefficient	Standard Error	t-statistic	p-value
omega	-0.0376	0.0169	-2.220	0.0264
alpha[1]	0.1724	0.0288	5.979	0.0000
gamma[1]	-0.0074	0.0093	-0.790	0.430
beta[1]	0.9863	0.0050	198.483	0.000

Diagnostic Checking of the Volatility Model

Post-estimation diagnostic tests were performed to verify the model. The Ljung-Box test yields a test statistic of 22.6343 (p = 0.0122). While this suggests that autocorrelation remains in the standardised residuals at the 5% significance level, it is common for high-frequency financial data. Given that the study's focus is on variance instead of mean prediction, the model is considered acceptable.

The ARCH-LM test yielded a statistic of 19.1311 (p = 0.0386). It indicates that ARCH effects remain unaccounted for by the EGARCH(1,1) model at the 5% significance level. Nevertheless, the minimised AIC and BIC suggested that the model represents the optimal feasible specification. Moreover, the highly significant ARCH and GARCH coefficients confirm that the model captures the majority of the volatility dynamics necessary for the subsequent analysis.

Trend Analysis of MYR/USD Conditional Volatility

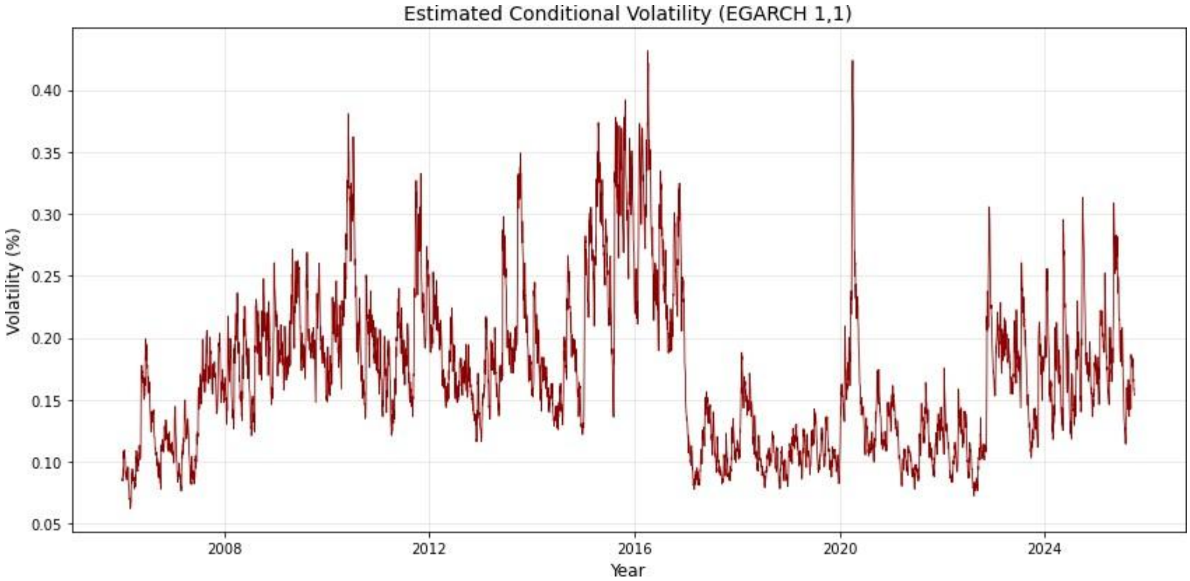
The descriptive statistics for conditional volatility are displayed in Table 4.1.4. The mean daily volatility is 0.1714% with a standard deviation of 0.0626%. The minimum conditional volatility during the observation period is 0.0624%, while the maximum is 0.4315%.

Table 4.1.4: Descriptive statistics for the conditional volatility of the MYR/USD exchange rate

Count	4844.0000
Mean	0.1714
Standard Deviation	0.0626
Min	0.0624
25%	0.1212
50%	0.1642
75%	0.2040
Max	0.4315

Figure 4.1.5 exhibits the graphical plot of the conditional volatility from January 2006 to September 2025. There are four distinct episodes of heightened risk. Firstly, the conditional volatility jumps in mid-2007, when the world was facing the prelude of the global financial crisis (Weinberg, n.d.). Secondly, the volatility increases again between 2014 and 2017 when the global oil price collapses (Stocker et al., 2024). At the same time, the 1MDB scandal emerged, and political instability in Malaysia heightened market uncertainty (Choudhury, 2015). Thirdly, a sharp spike occurred in early 2020 due to the COVID-19 outbreak (Northwestern Medicine, 2025). Lastly, volatility remains elevated from 2022 to 2025, driven by the aggressive monetary tightening by the U.S. Federal Reserve (Tepper, 2025).

Figure 4.1.5: Estimated conditional volatility from January 2006 to September 2025



Appendix 4.2

Stationarity Tests

Augmented Dickey-Fuller (ADF) Test Results

Table 4.2.1 displays the results of the ADF test. The results show that only EG and VOL are stationary at 5% significance level at the level form. The other variables become stationary after taking the first difference. This implies that EG and VOL are integrated of order zero, and the rest are integrated of order one.

Table 4.2.1: ADF Test Results

Variable	ADF Level (t-stat)	ADF Level (p-value)	ADF Diff (t-stat)	ADF Diff (p-value)	Result
EG	-3.338	0.013	-5.696	0.000	I(0)
VOL	-3.102	0.026	-12.292	0.000	I(0)
L_REER	-1.430	0.568	-12.355	0.000	I(1)
L_WPI	-1.092	0.718	-10.616	0.000	I(1)
L_CPI	-0.854	0.803	-10.412	0.000	I(1)
L_IPI	0.276	0.976	-5.371	0.000	I(1)

Phillips-Perron (PP) Test Results

Similarly, Table 4.2.2 presents the outcomes of the PP test. At the level form, only EG and

VOL are stationary, while the other variables become stationary in first differences. Therefore, both the ADF and PP tests provide consistent results.

Table 4.2.2: PP Test Results

Variable	ADF Level (t-stat)	ADF Level (p-value)	ADF Diff (t-stat)	ADF Diff (p-value)	Result
EG	-5.808	0.000	-24.264	0.000	I(0)
VOL	-4.226	0.001	-17.172	0.000	I(0)
L_REER	-1.135	0.701	-12.077	0.000	I(1)
L_WPI	-1.083	0.722	-10.861	0.000	I(1)
L_CPI	-1.351	0.606	-10.088	0.000	I(1)
L_IPI	-1.494	0.536	-38.511	0.000	I(1)

Appendix 4.3

Diagnostic Checking

Multicollinearity Test

Table 4.3.1 shows the correlation matrix, and Table 4.3.2 shows the Variance Inflation Factor (VIF). Several control variables, such as CPI, WPI, and IPI, are highly correlated (VIF > 10). However, it is acceptable in the time-series ARDL context as some studies have stated that multicollinearity is common and not considered as a model-breaking issue (Zainodin et al., 2011; Morrissey & Ruxton, 2018; Chan et al., 2022). According to Kumari (2008), macroeconomic variables often move together due to a shared business cycle. It is challenging to solve the problem by removing any variables, since all control variables are theoretically important. While WPI and IPI are highly correlated, both are retained to ensure the model controls for the demand and supply sides for export performance, preventing omitted variable bias from affecting the main estimator, VOL. Additionally, the VOL has a VIF of 1.22, indicating that multicollinearity does not affect the central hypothesis.

Table 4.3.1: Correlation Matrix

	VOL	L_REER	L_WPI	L_CPI	L_IPI
VOL	1				
L_REER	0.1510	1			
L_WPI	-0.2474	-0.7844	1		

L_CPI	-0.0986	-0.8511	0.9111	1	
L_IPI	-0.2211	-0.8451	0.9572	0.9399	1

Table 4.3.2: Variance Inflation Factor

VOL	1.2227
L_REER	4.0684
L_WPI	13.0912
L_CPI	11.1869
L_IPI	19.9595

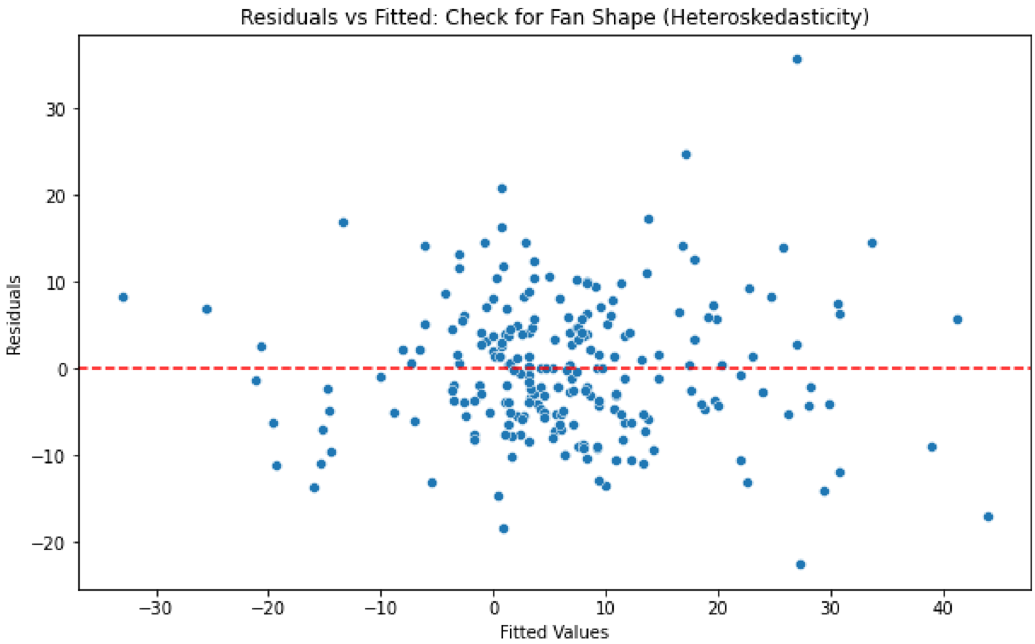
Autocorrelation Test

The Durbin-Watson statistic is 2.0007, approximate to 2. This suggests absence of serial correlation at the first-level. Moreover, the BG test at lag 2 has a p-value of 0.6913, indicating no autocorrelation problem. Hence, we concluded that the residuals are independent over time.

Heteroskedasticity Test

The *p*-value of the Breusch-Pagan Test is 0.0439. This is a marginal case of heteroscedasticity at the 5% significance level. The scatter plot shown in Figure 4.3.3 also indicates a fan-shaped pattern that reinforces the conclusion. Heteroskedasticity is very common in macroeconomic time-series data, driven by structural changes, external shocks, and policy transmission (Birau, 2012; Ahmed et al., 2016; Shi et al., 2018; Bruns & Lütkepohl, 2023). The sample periods cover major crises such as the global financial crisis, the oil price shock, and the COVID-19 pandemic, which may contribute to heteroskedastic errors. While the model suffers from heteroskedasticity, the coefficients remain linear and unbiased (Birau, 2012; Ahmed et al., 2016). However, it can also be concluded that the model is free of heteroskedasticity at the 1% significance level. Therefore, heteroskedasticity is acceptable, provided robust standard errors are applied.

Figure 4.3.3: Scatter Plot of Residuals against Fitted Values



Normality Test

The histograms of residuals and Q-Q plot are presented in Figures 4.3.4 and 4.3.5. The Jarque-Bera test has a *p*-value of 0.000, demonstrating that the data are not sampled from a normal distribution. The skewness is 0.5571, and the kurtosis is 4.4742. The data is slightly right-skewed with a fat tail. According to the Central Limit Theorem (CLT), the samples' mean will be normally distributed if the sample size is sufficiently large, even though the original variables are not normally distributed (Zygmont, 2023; Arachige, 2025). The study consists of 237 observations, suggesting that the violation of the normality assumption does not undermine the validity of the ARDL estimation. Besides, it is often acknowledged that macroeconomic variables typically exhibit non-normality, usually affected by crises and structural breaks (Naifar & Dohaiman, 2013; Wu, 2016). Therefore, we will use Student's t-distribution in our research to comply with the fat-tailed characteristic.

Figure 4.3.4: Histogram of Residuals

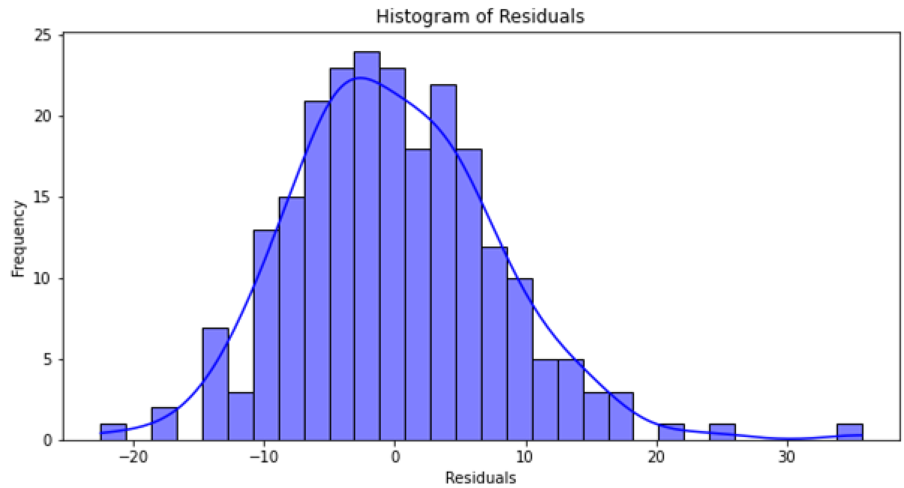
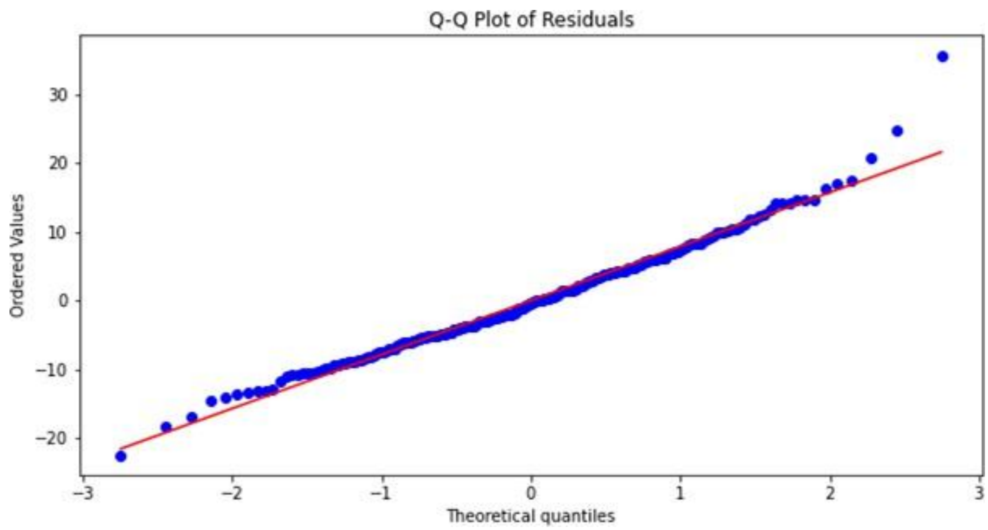


Figure 4.3.5: Q-Q Plot of Residuals



Appendix 4.4

Table 4.3: Short-Run (ECM) Estimation Results

Dependent Variable	D_EG	Log Likelihood		-803.29
Model	OLS	AIC		1639
Method	Least Squares	BIC		1694
Observations	231	F-statistic		7.618
R-squared	0.347	Prob(F-statistic)		0.000
Adjusted R-squared	0.301	Durbin-Watson		2.012

	Coefficient	Standard Error	t	p-value
const	0.0248	0.681	0.036	0.971
ECT_Lag1	-0.2918	0.054	-5.425	0.000
D_EG.L1	-0.2157	0.069	-3.115	0.002

D_EG.L2	0.0181	0.068	0.264	0.792
D_EG.L3	0.1126	0.069	1.634	0.104
D_EG.L4	0.0180	0.066	0.272	0.786
D_EG.L5	0.0285	0.062	0.463	0.644
D_VOL	-28.1536	19.082	-1.475	0.142
D_L_REER	-13.4441	46.454	-0.289	0.773
D_L_WPI	103.9833	64.403	1.615	0.108
D_L_CPI	296.3207	146.377	2.024	0.044
D_L_CPI.L1	178.2924	143.459	1.243	0.215
D_L_CPI.L2	14.4512	141.959	0.102	0.919
D_L_CPI.L3	314.5613	136.163	2.310	0.022
D_L_IPI	30.0177	14.092	2.130	0.034
D_L_IPI.L1	36.4224	12.731	2.861	0.005

Appendix 4.5

Raw Unrestricted ARDL Model Results

Dependent Variable	EG	Log Likelihood	-804.230
Model	ARDL(6,0,0,0,4,2)	S.D. of innovations	7.866
Method	Conditional MLE	AIC	1646.460
Observations	237	BIC	1711.866
R-squared	0.6692	HQIC	1672.841
Adjusted R-squared	0.6428	F-statistic	25.3487
Durbin-Watson	2.0007	Prob(F-statistic)	0.000

	Coefficient	Standard Error	z	P > |z|
const	59.6600	115.504	0.517	0.606
EG.L1	0.4891	0.067	7.334	0.000
EG.L2	0.2358	0.075	3.147	0.002
EG.L3	0.1009	0.075	1.353	0.178
EG.L4	-0.1067	0.074	-1.444	0.150
EG.L5	0.0075	0.073	0.103	0.918
EG.L6	-0.0293	0.063	-0.465	0.642
VOL.L0	-19.5841	10.724	-1.826	0.069
L_REER.L0	-21.7223	14.951	-1.453	0.148
L_WPI.L0	29.9109	32.663	0.916	0.361
L_CPI.L0	379.8211	136.339	2.786	0.006
L_CPI.L1	-210.6509	223.381	-0.943	0.347
L_CPI.L2	-162.2351	235.212	-0.690	0.491
L_CPI.L3	314.3203	226.875	1.385	0.167
L_CPI.L4	-320.4216	138.840	-2.308	0.022
L_IPI.L0	39.5599	14.829	2.668	0.008
L_IPI.L1	-18.5958	14.030	-1.325	0.186
L_IPI.L2	-42.1069	13.406	-3.141	0.002